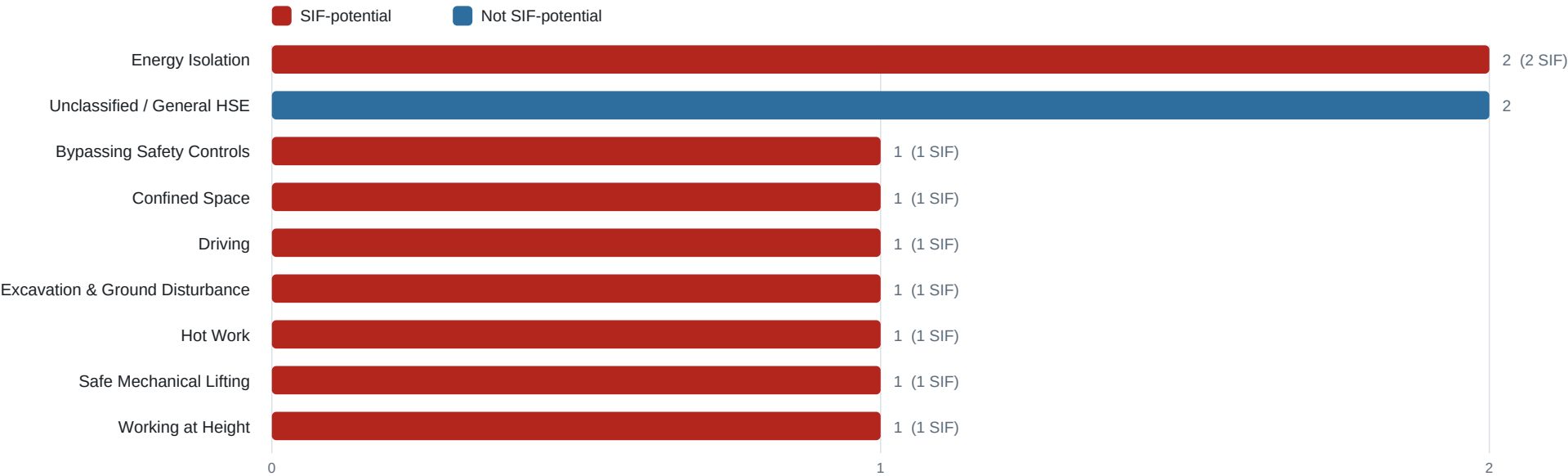


UA/UC & Near-Miss Analysis Report

Dataset: 11 field reports - 5 seed incidents and 6 batch-imported CSV rows. Each narrative was parsed by the SIF engine and classified against the IOGP Life-Saving Rules. A report is flagged SIF-potential only where a high-energy source is present **and** a critical barrier failed.

TOTAL REPORTS PROCESSED	SIF-POTENTIAL EVENTS	% SIF-POTENTIAL	TOP SIF EXPOSURE
11	9	81.8%	Energy Isolation

Exposure profile by IOGP rule



Key findings

- **9 of 11 reports (81.8%) satisfy both SIF conditions** - a high-energy source was present and a critical barrier had already failed. On this sample the precursor rate is the headline number, not the injury count: none of these reports records an actual injury.
- **Permit to work / JSA absent, expired or not followed is the most breached control** (3 reports), which makes work authorisation the highest-leverage intervention before any equipment-level fix.
- **Dominant energies: Electrical energy (2), Gravity / Fall from height (2).** Energy Isolation is the only rule to appear twice with both instances flagged, so electrical isolation discipline carries the largest share of the fatality exposure in this batch.
- **The 2 non-SIF reports are genuine low-energy observations** (housekeeping and walkway integrity), and both fall in the 2 reports that mapped to no Life-Saving Rule - no SIF signal was lost to the fallback. Mean confidence 0.74; the weakest extraction (0.14) is the sparsest narrative.

Where the controls broke, and against what energy

Barrier families are counted individually, so a report that breached two barriers appears against both.

Failed critical barrier	n	High-energy source present	n
Permit to work / JSA absent, expired or not followed	3	Electrical energy	2
Energy isolation / LOTO not applied or verified	2	Gravity / Fall from height	2
Housekeeping / walkway integrity lapse	2	Excavation / Ground collapse	1
Exclusion zone / barricading absent	1	Fire / Explosion	1
Fall protection not used / not anchored	1	Pressure / Stored energy	1
Fire prevention controls not in place	1	Process safety / Loss of containment	1
Gas testing / ventilation / atmospheric control missing	1	Suspended load / Mechanical	1
Safety device bypassed, inhibited or removed	1	Toxic / Asphyxiant atmosphere	1
Traffic / journey management control breached	1	Vehicle / Traffic motion	1

Method and limitations

Decision rule. SIF-potential = high-energy source present AND critical barrier failed. High energy without a barrier lapse is controlled work; a barrier lapse without high energy is a housekeeping or quality issue. The intersection is where fatalities originate, which is why both conditions are required rather than either alone.

Coverage. Nine IOGP Life-Saving Rules plus two upstream categories (Excavation & Ground Disturbance, Well Control & Process Containment); nine high-energy source families; twelve critical-barrier families. Activity, location and barrier text are extracted by keyword maps and pattern matching over the narrative.

Confidence. A 0.00-1.00 blend of rule-cue strength, energy cues and barrier cues. It measures how much evidence the text supplied, not how severe the event was. Every field degrades to an explicit fallback (Unclassified / General HSE, Unspecified activity, Location not stated, No barrier failure identified) rather than guessing.

Limitations. This is a deterministic heuristic prototype, not a trained classifier. It reads only what the narrative states: an unrecorded barrier failure cannot be detected, and terse or non-standard phrasing lowers confidence. In this dataset the two non-SIF reports (2 of 11) are genuine low-energy housekeeping observations, correctly separated from the 9 SIF precursors. Output supports human review by an HSE professional; it does not replace investigation or statutory reporting.

Parsed incident matrix

All 11 reports, in processing order. Severity mix: High 9, Low 2. Mean extraction confidence 0.74.

#	Source	SIF	IOGP Rule	Activity	Location	Barrier Failure	Energy Source	Sev.	Conf.
1	Seed incident	YES	Energy Isolation	Equipment maintenance	Pump station	Energy isolation / LOTO not applied or verified; Permit to work / JSA absent, expired or not followed	Electrical energy	High	1.00
2	Seed incident	YES	Working at Height	Component replacement / repair	Group Gathering Station (GGS)	Fall protection not used / not anchored	Gravity / Fall from height	High	0.90
3	Seed incident	YES	Safe Mechanical Lifting	Mechanical lifting operation	Well site / drilling location	Exclusion zone / barricading absent	Suspended load / Mechanical	High	1.00
4	Seed incident	no	Unclassified / General HSE	Inspection / condition monitoring	Oil Collecting Station (OCS)	Housekeeping / walkway integrity lapse	No high-energy source identified	Low	0.34
5	Seed incident	YES	Confined Space	Confined space entry	Effluent treatment plant	Gas testing / ventilation / atmospheric control missing	Pressure / Stored energy + Toxic / Asphyxiant atmosphere	High	0.80
6	CSV batch UAUC-2401	YES	Hot Work	Welding / grinding (hot work)	LPG plant	Fire prevention controls not in place	Fire / Explosion	High	0.80
7	CSV batch UAUC-2402	YES	Bypassing Safety Controls	Machining / workshop operation	Workshop / fabrication yard	Safety device bypassed, inhibited or removed	Process safety / Loss of containment	High	0.60
8	CSV batch UAUC-2403	YES	Excavation & Ground Disturbance	Excavation / ground disturbance	Pipeline right-of-way	Permit to work / JSA absent, expired or not followed	Gravity / Fall from height + Excavation / Ground collapse	High	0.90
9	CSV batch UAUC-2404	YES	Driving	Vehicle movement / transport	Access / field road	Traffic / journey management control breached	Vehicle / Traffic motion	High	0.70
10	CSV batch UAUC-2405	no	Unclassified / General HSE	Housekeeping / cleaning	Camp / administrative area	Housekeeping / walkway integrity lapse	No high-energy source identified	Low	0.14
11	CSV batch UAUC-2406	YES	Energy Isolation	Inspection / condition monitoring	Electrical substation / MCC room	Energy isolation / LOTO not applied or verified; Permit to work / JSA absent, expired or not followed	Electrical energy	High	1.00

Incident detail and extraction evidence

Source narrative with the fields the engine extracted from it, including the cues that drove each decision.

Incident 01 · Seed incident		SIF-POTENTIAL · confidence 1.00
Narrative	Near miss at Pump Station No. 3, Duliajan: during preventive maintenance of the booster pump the 11 kV feeder cable was left ungrounded and the breaker was not racked out. The electrician started cable jointing without LOTO applied and no isolation certificate was raised.	
IOGP rule	Energy Isolation	
Activity	Equipment maintenance	
Location	Pump station	
Barrier failure	Energy isolation / LOTO not applied or verified; Permit to work / JSA absent, expired or not followed	
Energy source	Electrical energy	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: 11 kv, breaker, isolation, loto; energy cues: 11 kv, breaker, kv; barrier cues: left ungrounded, no isolation certificate, no isolation certificate was , ungrounded	

Incident 02 · Seed incident		SIF-POTENTIAL · confidence 0.90
Narrative	Unsafe act observed at GGS-4 tank farm: a contract worker was standing on the incomplete scaffold at about 6 metres near the crude tank roof to replace a light fitting. He was wearing a full-body harness but the lanyard was not anchored and the guardrail on the working platform was missing.	
IOGP rule	Working at Height	
Activity	Component replacement / repair	
Location	Group Gathering Station (GGS)	
Barrier failure	Fall protection not used / not anchored	
Energy source	Gravity / Fall from height	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: guardrail, harness, lanyard, roof; energy cues: 6 metres, roof, scaffold; barrier cues: incomplete scaffold, lanyard was not anchored	

Incident 03 · Seed incident		SIF-POTENTIAL · confidence 1.00
Narrative	Unsafe condition during casing shifting at the drill site: the rigging crew and a helper stood directly under the suspended load while the hydra slewed the casing bundle. The area was not barricaded, there was no banksman, and one sling was found frayed with an expired third-party certificate.	
IOGP rule	Safe Mechanical Lifting	
Activity	Mechanical lifting operation	
Location	Well site / drilling location	
Barrier failure	Exclusion zone / barricading absent	
Energy source	Suspended load / Mechanical	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: hydra, rigging, slewed, sling; energy cues: hydra, sling, suspended load; barrier cues: area was not barricaded, no banksman, stood directly under, under the suspended load	

Incident 04 · Seed incident		NOT SIF-POTENTIAL · confidence 0.34
Narrative	Unsafe condition reported in the OCS process area: minor oil spillage on the walkway near the manifold made the chequered plate slippery during routine rounds. Poor housekeeping, no injury; area was cleaned and anti-skid tape applied the same shift.	
IOGP rule	Unclassified / General HSE	
Activity	Inspection / condition monitoring	
Location	Oil Collecting Station (OCS)	
Barrier failure	Housekeeping / walkway integrity lapse	
Energy source	No high-energy source identified	
Decision	high energy = no; barrier failed = yes; severity = Low	
Matched cues	barrier cues: oil spillage, poor housekeeping, slippery	
Incident 05 · Seed incident		SIF-POTENTIAL · confidence 0.80
Narrative	Near miss during vessel entry at the Effluent Treatment Plant: two workers entered the separator sump for cleaning without gas testing being carried out and without forced ventilation. H2S was later detected at the manway and there was no hole watch posted; the permit had expired the previous shift.	
IOGP rule	Confined Space	
Activity	Confined space entry	
Location	Effluent treatment plant	
Barrier failure	Gas testing / ventilation / atmospheric control missing	
Energy source	Pressure / Stored energy + Toxic / Asphyxiant atmosphere	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: gas testing, h2s, hole watch, ventilation; energy cues: h2s, separator; barrier cues: no hole watch, without forced ventilation	
Incident 06 · CSV batch · UAUC-2401		SIF-POTENTIAL · confidence 0.80
Narrative	Unsafe act at the LPG bottling plant: a fitter used an angle grinder next to the filling carousel without a hot work permit and with no fire watch posted; flammable vapour was present at the filling point.	
IOGP rule	Hot Work	
Activity	Welding / grinding (hot work)	
Location	LPG plant	
Barrier failure	Fire prevention controls not in place	
Energy source	Fire / Explosion	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: fire watch, grinder, hot work; energy cues: fire, flammable, grinder, hot work; barrier cues: no fire watch	

Incident 07 · CSV batch · UAUC-2402		SIF-POTENTIAL · confidence 0.60
Narrative	Unsafe condition at the workshop: the lathe machine guard was removed and the operator continued turning a shaft; interlock was bypassed with a jumper.	
IOGP rule	Bypassing Safety Controls	
Activity	Machining / workshop operation	
Location	Workshop / fabrication yard	
Barrier failure	Safety device bypassed, inhibited or removed	
Energy source	Process safety / Loss of containment	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: bypassed, interlock, jumper, machine guard; barrier cues: guard was removed, machine guard was removed; energy inferred from rule 'Bypassing Safety Controls'	
Incident 08 · CSV batch · UAUC-2403		SIF-POTENTIAL · confidence 0.90
Narrative	Near miss on the pipeline right-of-way: the excavator struck a buried cable while trenching to 2.5 m depth; no ground disturbance permit and no cable route survey had been done.	
IOGP rule	Excavation & Ground Disturbance	
Activity	Excavation / ground disturbance	
Location	Pipeline right-of-way	
Barrier failure	Permit to work / JSA absent, expired or not followed	
Energy source	Gravity / Fall from height + Excavation / Ground collapse	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: buried cable, excavator, trenching; energy cues: 5 m, buried cable, excavator, trenching; barrier cues: no cable route survey had been done, no ground disturbance permit	
Incident 09 · CSV batch · UAUC-2404		SIF-POTENTIAL · confidence 0.70
Narrative	Unsafe act during transport: the tanker driver was overspeeding on the field approach road and was not wearing a seat belt; journey management plan was not raised.	
IOGP rule	Driving	
Activity	Vehicle movement / transport	
Location	Access / field road	
Barrier failure	Traffic / journey management control breached	
Energy source	Vehicle / Traffic motion	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: driver, journey management, overspeeding, seat belt; energy cues: overspeeding, tanker; barrier cues: overspeeding	

Incident 10 · CSV batch · UAUC-2405		NOT SIF-POTENTIAL · confidence 0.14
Narrative	Unsafe condition at the camp canteen: a loose chequered plate near the entrance created a trip hazard. Minor housekeeping issue, no injury.	
IOGP rule	Unclassified / General HSE	
Activity	Housekeeping / cleaning	
Location	Camp / administrative area	
Barrier failure	Housekeeping / walkway integrity lapse	
Energy source	No high-energy source identified	
Decision	high energy = no; barrier failed = yes; severity = Low	
Matched cues	barrier cues: loose chequered plate	
Incident 11 · CSV batch · UAUC-2406		SIF-POTENTIAL · confidence 1.00
Narrative	Near miss at the substation: the 33 kV switchgear panel door was left open with the busbar still live while the technician took thermography readings; the panel had not been de-energised and no isolation certificate was available.	
IOGP rule	Energy Isolation	
Activity	Inspection / condition monitoring	
Location	Electrical substation / MCC room	
Barrier failure	Energy isolation / LOTO not applied or verified; Permit to work / JSA absent, expired or not followed	
Energy source	Electrical energy	
Decision	high energy = yes; barrier failed = yes; severity = High	
Matched cues	rule cues: 33 kv, busbar, de-energised, isolation; energy cues: 33 kv, busbar, energised, kv; barrier cues: no isolation certificate, no isolation certificate was available, still live	